In [8]:

```python
import pandas as pd
df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-material\WHR2023.csv"

print("Original columns:", df.columns.tolist())

# Select correct columns explicitly
df_clean = df[[
    'Country name',
    'Ladder score',
    'Logged GDP per capita',
    'Social support',
    'Healthy life expectancy',
    'Freedom to make life choices',
    'Generosity',
    'Perceptions of corruption'
]].copy()

# Rename properly
df_clean.columns = [
    'Country',
    'Happiness_Score',
    'GDP',
    'Social_Support',
    'Life_Expectancy',
    'Freedom',
    'Generosity',
    'Corruption'
]

print(df_clean.head())
```

```
Original columns: ['Country name', 'Ladder score', 'Standard error of ladder score', 'uppe
rwhisker', 'lowerwhisker', 'Logged GDP per capita', 'Social support', 'Healthy life expect
ancy', 'Freedom to make life choices', 'Generosity', 'Perceptions of corruption', 'Ladder
score in Dystopia', 'Explained by: Log GDP per capita', 'Explained by: Social support', 'E
xplained by: Healthy life expectancy', 'Explained by: Freedom to make life choices', 'Expl
ained by: Generosity', 'Explained by: Perceptions of corruption', 'Dystopia + residual']
Country   Happiness_Score      GDP   Social_Support   Life_Expectancy   \ 0         Finland
7.804  10.792           0.969           71.150
1       Denmark           7.586  10.962           0.954           71.250
2       Iceland           7.530  10.896           0.983           72.050
3       Israel           7.473  10.639           0.943           72.697
4       Netherlands           7.403  10.942           0.930           71.550

   Freedom  Generosity  Corruption  0
0.961      -0.019      0.182
1    0.934       0.134       0.196
2    0.936       0.211       0.668
```

In

```
3    0.809      -0.023       0.708
4    0.887       0.213       0.379
```

[10]:

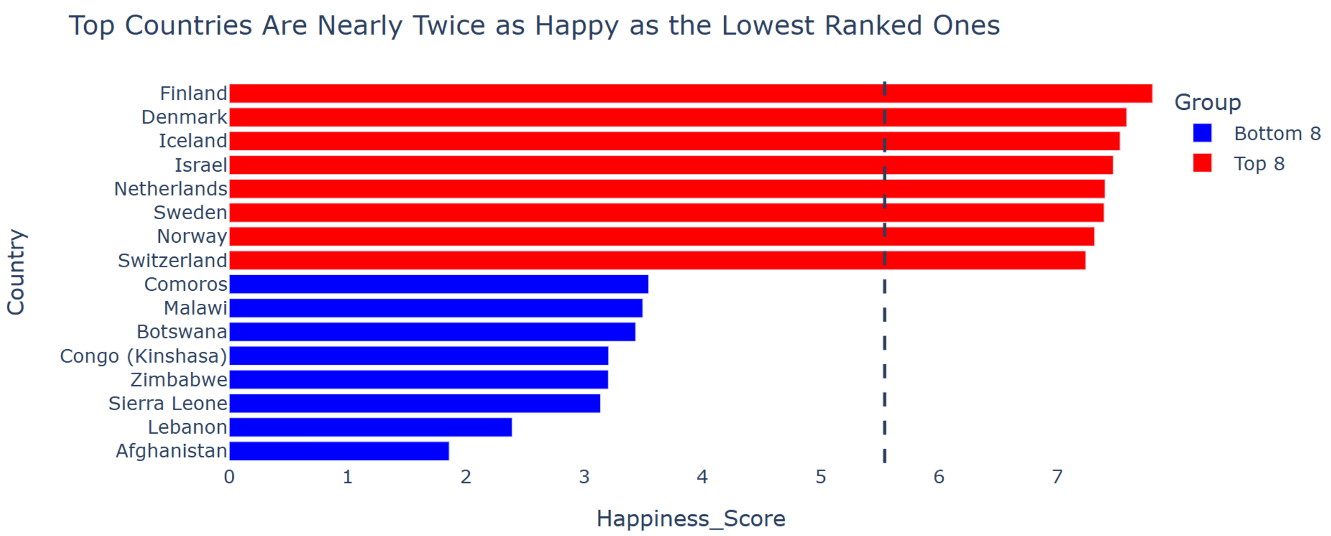
```
df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-material\WHR2023.csv"
print("Original columns:", df.columns.tolist())

df_subset = df.iloc[:, :9]  # Select only the first 9 columns
df_subset.columns =
['Country','Region','Happiness_Score','GDP','Social_Support',
'Life_Expectancy','Freedom','Generosity','Corruption']

print(f"Dataset: {len(df_subset)} countries, {len(df_subset.columns)} columns")
print(df_subset.head())
```

```
Original columns: ['Country name', 'Ladder score', 'Standard error of ladder score', 'uppe
rwhisker', 'lowerwhisker', 'Logged GDP per capita', 'Social support', 'Healthy life expect
ancy', 'Freedom to make life choices', 'Generosity', 'Perceptions of corruption', 'Ladder
score in Dystopia', 'Explained by: Log GDP per capita', 'Explained by: Social support', 'E
xplained by: Healthy life expectancy', 'Explained by: Freedom to make life choices', 'Expl
ained by: Generosity', 'Explained by: Perceptions of corruption', 'Dystopia + residual']
Dataset: 137 countries, 9 columns
       Country  Region  Happiness_Score    GDP  Social_Support  \ 0
Finland   7.804           0.036  7.875           7.733
1       Denmark   7.586           0.041  7.667           7.506
2       Iceland   7.530           0.049  7.625           7.434
3       Israel   7.473           0.032  7.535           7.411
4       Netherlands   7.403           0.029  7.460           7.346

   Life_Expectancy  Freedom  Generosity  Corruption  0
10.792    0.969      71.150       0.961
1           10.962    0.954      71.250       0.934
2           10.896    0.983      72.050       0.936
3           10.639    0.943      72.697       0.809
4           10.942    0.930      71.550       0.887
```

In

```
region_map = {
    'Finland': 'Western Europe',
    'Denmark': 'Western Europe',
    'Iceland': 'Western Europe',
    'Netherlands': 'Western Europe',
    'Sweden': 'Western Europe',
    'Norway': 'Western Europe',
    'Germany': 'Western Europe',
    'France': 'Western Europe',
    'Spain': 'Western Europe',

    'India': 'South Asia',
    'Pakistan': 'South Asia',
    'Bangladesh': 'South Asia',

    'China': 'East Asia',
    'Japan': 'East Asia',
    'South Korea': 'East Asia',

    'Nigeria': 'Sub-Saharan Africa',
    'Kenya': 'Sub-Saharan Africa',
    'Ethiopia': 'Sub-Saharan Africa',
    'Congo (Kinshasa)': 'Sub-Saharan Africa',
    'Sierra Leone': 'Sub-Saharan Africa',

    'Brazil': 'Latin America',
    'Argentina': 'Latin America',
    'Mexico': 'Latin America'
}

df_clean['Region'] = df_clean['Country'].map(region_map)
df_clean = df_clean.dropna(subset=['Region'])
```

In [11]:



In

```python
import pandas as pd
import plotly.express as px   # <-- add this

region_avg = df_clean.groupby('Region')['Happiness_Score'].mean().reset_index()
region_avg = region_avg.sort_values('Happiness_Score')

fig1 = px.bar(
    region_avg,
    x='Happiness_Score',
    y='Region',
    orientation='h'
)

fig1.update_xaxes(range=[0, region_avg['Happiness_Score'].max()])

fig1.update_layout(
    title="Western Europe Shows the Highest Happiness Levels Among Selected Regions" ,
    plot_bgcolor='white'
)

fig1.show()

```

[20]:

In [17]:

```python
import plotly.express as px

# Top and Bottom
top8 = df_clean.nlargest(8, 'Happiness_Score').copy()
top8['Group'] = 'Top 8'

bottom8 = df_clean.nsmallest(8, 'Happiness_Score').copy()
bottom8['Group'] = 'Bottom 8'

combined = pd.concat([bottom8, top8])
combined = combined.sort_values('Happiness_Score')

# Global average
global_avg = df_clean['Happiness_Score'].mean()

# Plot
fig = px.bar(
    combined,
    x='Happiness_Score',
    y='Country',
    orientation='h',
    color='Group',
    color_discrete_map={'Top 8': 'green', 'Bottom 8': 'red'}
)

fig.add_vline(x=global_avg, line_dash="dash")

fig.update_xaxes(range=[0, combined['Happiness_Score'].max()])

fig.update_layout(
    title="Top Countries Are Nearly Twice as Happy as the Lowest Ranked Ones",
    plot_bgcolor='white'
)

fig.show()

```